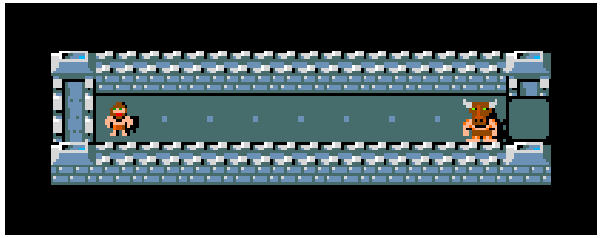
MiniHack: Wand of Death

A turn-based dungeon game: navigate the map and use a wand against a minotaur.

Player · Wand of death · Minotaur · Staircase

Easy

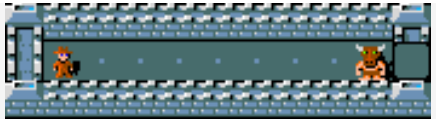
Default limit: **50 actions**



Recorded episode

Wand in inventory; sleeping minotaur.

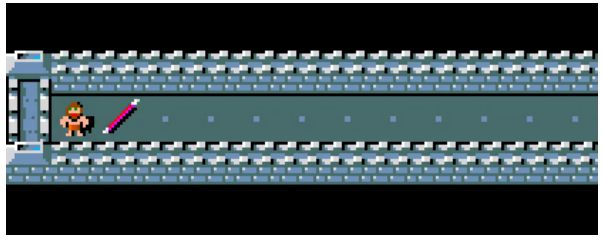
Goal: Kill the minotaur.



Full map

Medium

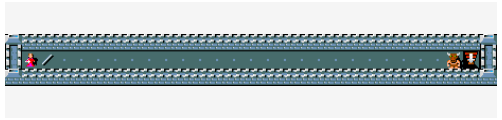
Default limit: **150 actions**



Recorded episode

Wand on the floor; sleeping minotaur.

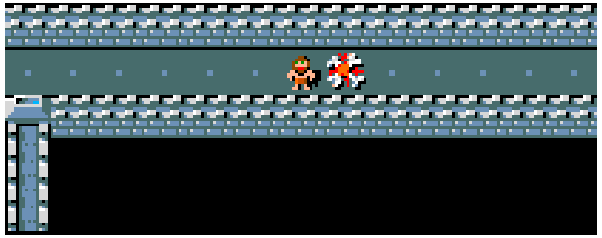
Goal: Reach the staircase.



Full map

Hard

Default limit: **400 actions**



Recorded episode

Find the wand; minotaur is awake.

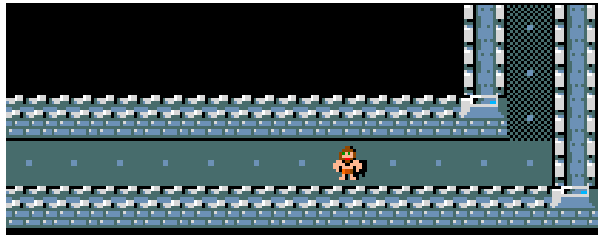
Goal: Reach the staircase.



Full map

Pro

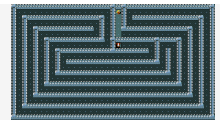
Default limit: **1,000 actions**



Excerpt · first 60 actions

Wand hidden in a maze; minotaur awake.

Goal: Reach the staircase.



Full map

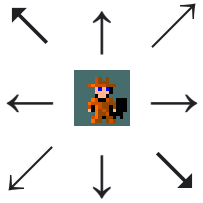
Fixed terrain per level; some placements vary in Hard and Pro. Full maps are revealed for illustration. Limits count all inputs, including menu replies.

The agent’s interface

Episodes end on success, death or the environment’s action limit.

Actions

11 inputs in the Restricted action set



8 directions
Move to a neighbouring square.

3 commands
Pick up · Use a wand · Fire

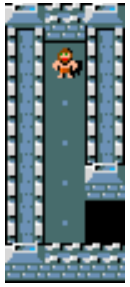
Using a wand

1. Choose “Use a wand”.
2. Select the wand from inventory.
3. Choose a firing direction.

Prompts reuse the same inputs: a direction can aim instead of move.

Observations

Visible objects and remembered terrain; unexplored areas remain hidden.



Pro at the start: a small part of the maze.

Inventory

Items the character is carrying.

Character state

Health, position and game time.

Messages and prompts

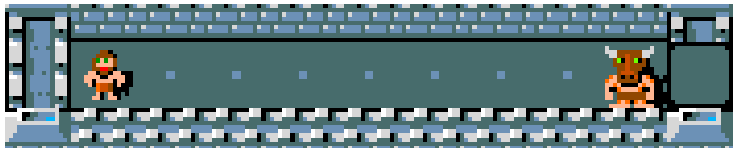
Feedback and requests for the next choice, e.g. “In what direction?”

Default rewards

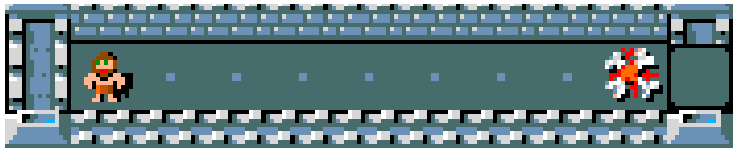
+1	**Success:** minotaur killed on Easy; staircase reached on other levels.
0	**No success,** including death or the action limit.
−0.01	**Extra cost** when game time does not advance, e.g. a menu reply or blocked move.

Easy: a successful wand attack

Before · minotaur alive



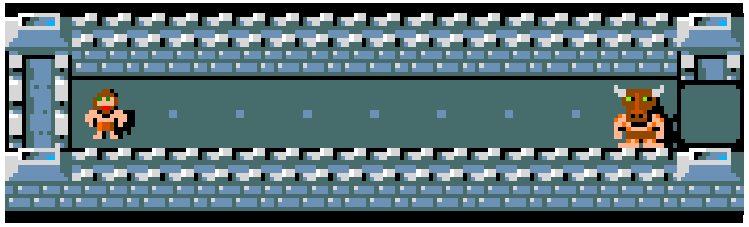
After · minotaur killed · **+1 reward**



Interactive demonstration

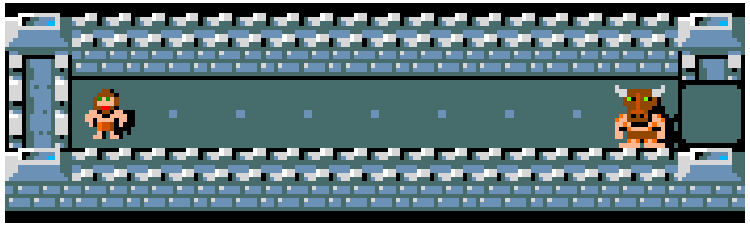
Play all four levels on the website, with optional hints.

1. Use wand R



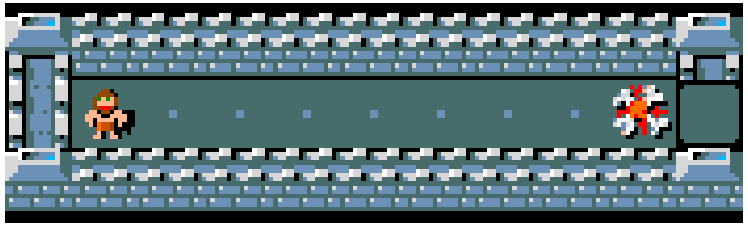
```
What do you want to zap? [f or ?*]
```

2. Select wand F



```
In what direction?
```

3. Aim east D



```
You kill the minotaur! Welcome to
experience level 2.
```

Starting inventory · Easy

NetHack item letters

- a a +1 club (weapon in hand)
- b a +2 sling (alternate weapon; not wielded)
- c 16 uncursed flint stones (in quiver pouch)
- d 30 uncursed rocks
- e an uncursed +0 leather armor (being worn)
- f a uranium wand

Here, “a uranium wand” is the unidentified wand of death.

Move / aim
WASD + Q/E/Z/C

Commands
P pick up · R use wand · F select / fire

Restart
Space · same starting setup